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CYBER SKYLINE

The National Cyber League
A Community Where Cybersecurity Is a Passion

Ji Fu Aaron Ng

NCL Spring 2025 Team Game Scouting Report

Dear Ji Fu Aaron Ng (Team "SANS.edu Sentinels 713"),

Thank you for participating in the National Cyber League (NCL) Spring 2025 Season! Our goal is to prepare the next generation of cybersecurity professionals, and your participation is helping achieve that goal.

The NCL was founded in May 2011 to provide an ongoing virtual training ground for collegiate students to develop, practice, and validate their cybersecurity skills in preparation for further learning, industry certifications, and career readiness. The NCL scenario-based challenges were designed around performance-based exam objectives of CompTIA certifications and are aligned to the National Initiative for Cybersecurity Education (NICE) Cybersecurity Workforce Framework published by the National Institute of Standards and Technology (NIST).

As you look to a future career in cybersecurity, we hope you find this report to be valuable in both validating skills and identifying areas for improvement across the nine NCL skills categories. You can use this NCL Scouting Report to:

- Validate your skills to employers in any job application or professional portfolio;
- Show case your achievements and strengths by including the Score Card view of your performance as part of your résumé or simply sharing the validation link so that others may view the detailed version of this report.

The NCL Spring 2025 Season had 9,216 students/players and 596 faculty/coaches from 510 two- and four-year schools & 288 high schools across all 50 U.S. states registered to play. The Individual Game Capture the Flag (CTF) event took place from April 11 through April 13. The Team Game CTF event took place from April 25 through April 27. The games were conducted in real-time for students across the country.

NCL is powered by Cyber Skyline's cloud-based skills evaluation platform. Cyber Skyline hosted the scenario-driven cybersecurity challenges for players to compete and track their progress in real-time.



To validate this report, please access: cyberskyline.com/report/EABQTTJRFQRX

Congratulations for your participation in the NCL Spring 2025 Team Game! We hope you will continue to develop your knowledge and skills and make meaningful contributions as part of the Information Security workforce!

Dr. David Zeichick
NCL Commissioner



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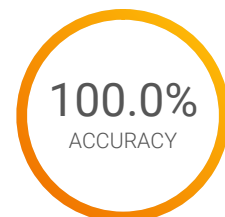
NATIONAL CYBER LEAGUE SCORE CARD

NCL SPRING 2025 TEAM GAME

YOUR TOP CATEGORIES



NATIONAL RANK
14TH PLACE
OUT OF 4779
PERCENTILE
100TH



Average: 55.4%

[cyberskyline.com/report](https://cyberskyline.com/report/EABQTTJRFQRX)
ID: EABQTTJRFQRX

Learn more at nationalcyberleague.org



NCL Spring 2025 Team Game

The NCL Team Game is designed for student players nationwide to compete in realtime in the categories listed below. The Team Game promotes camaraderie and evaluates the collective technical cybersecurity skills of the team members.

14TH PLACE OUT OF 4779 NATIONAL RANK	2860 POINTS OUT OF 3000 PERFORMANCE SCORE	100.0% ACCURACY	99.2% COMPLETION
100 th National Percentile	Average: 989.7 Points	Average: 55.4%	Average: 42.7%
Cryptography	325 POINTS OUT OF 325	100.0% ACCURACY	COMPLETION: 100.0%
Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.			
Enumeration & Exploitation	260 POINTS OUT OF 380	100.0% ACCURACY	COMPLETION: 87.5%
Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.			
Forensics	300 POINTS OUT OF 300	100.0% ACCURACY	COMPLETION: 100.0%
Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.			
Log Analysis	315 POINTS OUT OF 315	100.0% ACCURACY	COMPLETION: 100.0%
Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.			
Network Traffic Analysis	300 POINTS OUT OF 300	100.0% ACCURACY	COMPLETION: 100.0%
Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.			
Open Source Intelligence	350 POINTS OUT OF 350	100.0% ACCURACY	COMPLETION: 100.0%
Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.			
Password Cracking	320 POINTS OUT OF 320	100.0% ACCURACY	COMPLETION: 100.0%
Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.			
Scanning & Reconnaissance	300 POINTS OUT OF 300	100.0% ACCURACY	COMPLETION: 100.0%
Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.			
Web Application Exploitation	290 POINTS OUT OF 310	100.0% ACCURACY	COMPLETION: 100.0%
Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.			

Note: Survey module (100 points) was excluded from this report.



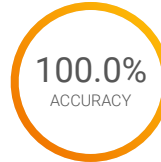


Cryptography Module

Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.

2ND PLACE
OUT OF 4779
NATIONAL RANK

325 POINTS
OUT OF 325
PERFORMANCE SCORE



Average: 58.9%



Average: 46.0%

TOP NICE WORKROLES

Security Control Assessor
Secure Software Assessor
Exploitation Analyst
Cyber Operator
Security Architect

100th National
Percentile

Average: 110.6 Points

So Basic (Easy)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: **100.0%**

Decode the plaintext of messages encoded with common number bases

For Fun (Easy)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: **100.0%**

Decrypt messages that are encrypted using different shift ciphers

Crypto Therapy (Medium)

50 POINTS
OUT OF 50

100.0%
ACCURACY

COMPLETION: **100.0%**

Exploit a poor implementation of RSA to decrypt sensitive data

Skip To My Lou (Medium)

60 POINTS
OUT OF 60

100.0%
ACCURACY

COMPLETION: **100.0%**

Decrypt a message that was encrypted using the Skipjack cipher

Infinite Cosmic Power (Hard)

125 POINTS
OUT OF 125

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform a differential power attack on a power trace to obtain an AES encryption key





Enumeration & Exploitation Module

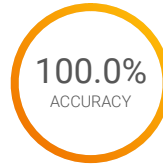
Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.

33 RD PLACE
OUT OF 4779
NATIONAL RANK

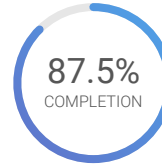
100th National
Percentile

260 POINTS
OUT OF 380
PERFORMANCE SCORE

Average: 132.2 Points



Average: 57.3%



Average: 50.1%

TOP NICE WORKROLES

Cyber Operator
Target Developer
Exploitation Analyst
Software Developer
Systems Security Analyst

Ransomed (Easy)

Analyze source code to bypass an XOR-based authentication mechanism

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Crawling (Medium)

Perform a privilege escalation attack by coercing a binary with elevated permissions into running untrusted software

150 POINTS
OUT OF 150

100.0%
ACCURACY

COMPLETION: **100.0%**

ExploitExhibit (Hard)

Exploit a C++ program by undermining the heap memory or leaking vtable pointers

10 POINTS
OUT OF 130

100.0%
ACCURACY

COMPLETION: **50.0%**

Forensics Module

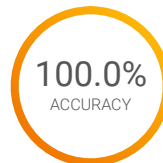
Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.

7 TH PLACE
OUT OF 4779
NATIONAL RANK

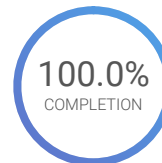
100th National
Percentile

300 POINTS
OUT OF 300
PERFORMANCE SCORE

Average: 136.7 Points



Average: 56.8%



Average: 48.1%

TOP NICE WORKROLES

Cyber Defense Forensics
Analyst
Cyber Crime Investigator
Cyber Defense Incident
Responder
Cyber Defense Analyst

Origin Story (Easy)

Investigate a Git repository to find leaked secrets

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Ransom Residue (Medium)

Perform a forensic investigation of a computer that was compromised by ransomware by analyzing a disk image file

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Archive (Hard)

Automate the extraction of files by matching file signatures in a raw byte stream

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**





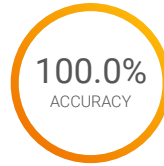
Log Analysis Module

Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.

2ND PLACE
OUT OF 4779
NATIONAL RANK

315 POINTS
OUT OF 315
PERFORMANCE SCORE

Average: 161.6 Points



Average: 52.2%



Average: 53.9%

TOP NICE WORKROLES

Cyber Defense Analyst
Systems Security Analyst
All-Source Analyst
Cyber Defense Forensics Analyst
Data Analyst

100th National
Percentile

What Remains (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze event logs from a Windows 7 workstation that was compromised by EternalBlue

Compromising (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a NetFlow log to identify compromised devices and malicious behavior

Coords (Hard)

115 POINTS
OUT OF 115

100.0%
ACCURACY

COMPLETION: **100.0%**

Parse a custom binary file format and perform geospatial analysis of drone flight paths

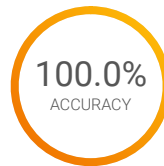
Network Traffic Analysis Module

Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.

3RD PLACE
OUT OF 4779
NATIONAL RANK

300 POINTS
OUT OF 300
PERFORMANCE SCORE

Average: 147.2 Points



Average: 60.6%



Average: 65.1%

TOP NICE WORKROLES

Cyber Defense Analyst
All-Source Analyst
Cyber Defense Incident Responder
Target Network Analyst
Cyber Operator

100th National
Percentile

HTTP Forever (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a packet capture with HTTP traffic to identify web browsing behavior

SQL Storage Secrets (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze PostgreSQL traffic to discover information about a database and the binary content inside it

Ping of Domin8tion (Hard)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a packet capture to extract an image that was exfiltrated by encoding data in ICMP packet headers





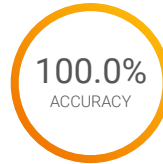
Open Source Intelligence Module

Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.

3RD PLACE
OUT OF 4779
NATIONAL RANK

350 POINTS
OUT OF 350
PERFORMANCE SCORE

Average: 181.0 Points



Average: 53.9%



Average: 64.6%

TOP NICE WORKROLES

Systems Security Analyst
Target Developer
System Administrator
Research & Development
Specialist
Cyber Intel Planner

100th National
Percentile

Rules of Conduct (Easy)

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Introductory challenge on acceptable conduct during NCL

Swap (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Utilize blockchain tools to perform an analysis of cryptocurrency transactions

Granny (Medium)

60 POINTS
OUT OF 60

100.0%
ACCURACY

COMPLETION: **100.0%**

Use open source tools to find information about the subject of an image

Sla.cking (Medium)

60 POINTS
OUT OF 60

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform introspection on deep web forums using the Internet Archive

Gramma Brasiliensis (Hard)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Conduct an open source investigation on an individual to create a timeline of their actions and cryptocurrency transactions



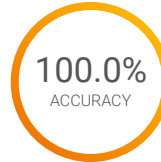


Password Cracking Module

Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.

11TH PLACE
OUT OF 4779
NATIONAL RANK

320 POINTS
OUT OF 320
PERFORMANCE SCORE



Average: 80.8%



Average: 53.7%

TOP NICE WORKROLES

Cyber Operator
Exploitation Analyst
Systems Security Analyst
Cyber Defense Incident
Responder
Cyber Crime Investigator

100th National
Percentile

Average: 151.0 Points

Loaded Hash Browns (Easy)

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Generate password hashes using MD5, SHA1, and SHA256

You're So Basic (Easy)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: **100.0%**

Crack MD5 password hashes for commonly used passwords

786b63642e636f6d2f393336 (Medium)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: **100.0%**

Crack Windows NTLM password hashes that may not be found in common rainbow tables

Zip (Medium)

50 POINTS
OUT OF 50

100.0%
ACCURACY

COMPLETION: **100.0%**

Crack the insecure password for a protected zip file

Crunch Time (Medium)

50 POINTS
OUT OF 50

100.0%
ACCURACY

COMPLETION: **100.0%**

Create a custom wordlist to crack passwords based on known password rules

Router (Hard)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Build a custom wordlist to crack passwords by augmenting permutation rules using known password complexity requirements





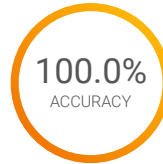
Scanning & Reconnaissance Module

Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.

15TH PLACE
OUT OF 4779
NATIONAL RANK

300 POINTS
OUT OF 300
PERFORMANCE SCORE

Average: 106.3 Points



Average: 46.8%



Average: 35.7%

TOP NICE WORKROLES

Vulnerability Assessment
Analyst
Target Network Analyst
Cyber Operations Planner
Target Developer
Security Control Assessor

100th National
Percentile

LuckyCat (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Classify malware from benign samples by creating custom Yara rules

Schrödinger's Port (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Disable a local firewall and perform a port scan on an internal network

Ephemeral (Hard)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Scan a device and perform a port knocking sequence to access a hidden service

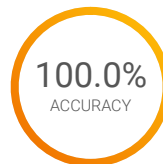
Web Application Exploitation Module

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.

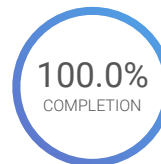
8TH PLACE
OUT OF 4779
NATIONAL RANK

290 POINTS
OUT OF 310
PERFORMANCE SCORE

Average: 96.3 Points



Average: 69.9%



Average: 35.9%

TOP NICE WORKROLES

Cyber Operator
Software Developer
Exploitation Analyst
Systems Security Analyst
Database Administrator

100th National
Percentile

Doohickey Divulgence (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform analysis on minified source code to decode and extract sensitive information

Pierre's Store (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform a SQL injection attack to leak sensitive information from a web application

Cash Rules Everything Around Me (Hard)

90 POINTS
OUT OF 110

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform a series of cache deception attacks against a web application

